

Nidhi Prashar

Chemistry



- **16+ Years of teaching experience**
- **Gold Medallist in the Master's Degree, UGC NET Qualified**
- **IJSO EXPERT with 150+ NSEJS Selections**
- **MENTORED IJSO GOLD MEDALISTS**
- **Teaching and mentoring students IIT, NEET, NSEC, IJSO exams**
- **100+ Selections in Science Olympiads**
- **Produced Single and double digit ranks in JEE/NEET (AIR 4 in AIIMS)**

Examination Schedule

NSEA (Astronomy)	Saturday 21.11.26	2.30 pm to 4.30 pm
NSEP (Physics)	Sunday 22.11.26	8.30 am to 10.30 am
NSEC (Chemistry)	Sunday 22.11.26	11.30 am to 1.30 pm
NSEB (Biology)	Sunday 22.11.26	2.30 pm to 4.30 pm
NSEJS (Junior Science)	Sunday 22.11.26	2.30 pm to 4.30 pm

Telegram Channel



ORGANIC CHEMISTRY

Historical Classification Of Chemical Compounds

CHEMICAL COMPOUNDS

```
graph TD; A[CHEMICAL COMPOUNDS] --> B[ORGANIC COMPOUNDS]; A --> C[INORGANIC COMPOUNDS];
```

The compounds obtained Directly or indirectly from living organism such as plants and animals were called 'ORGANIC COMPOUNDS'

Examples :
Oils, Sugar

The compounds obtained from non-living Sources were called 'INORGANIC COMPOUNDS'

Examples :
Proteins, Salt ,CaCO₃

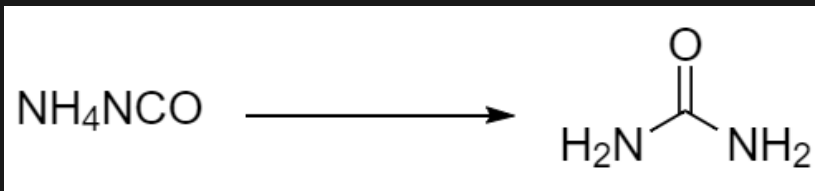
Theory of Vitalism

Berzellius



- There are some mysterious forces from nature which operates on plants and animals due to which they produces organic compounds.
- Since this forces can't be artificially prepared, organic compounds can't be prepared artificially .

Friedrich Wohler



Ammonium
cyanate

Urea
*first organic
compound*

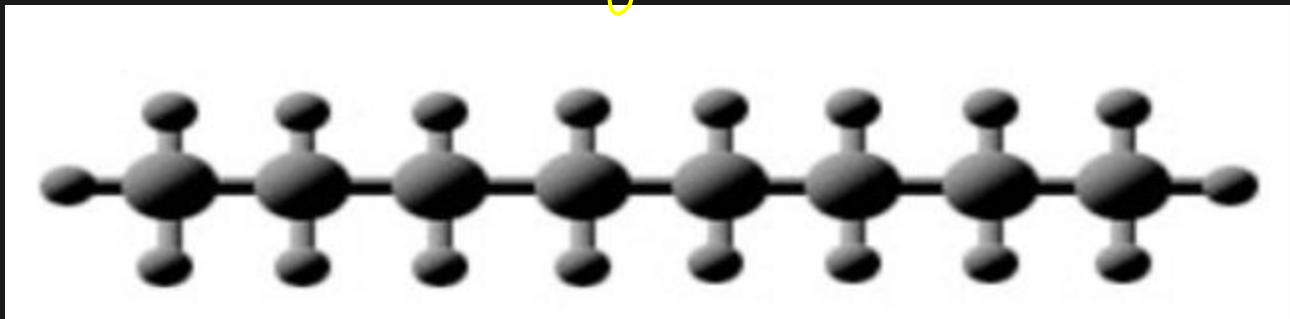
Synthesis of urea (from ammonium cyanate) was the first synthesis of an organic chemical compound from an inorganic one; it contributed to the foundation of modern organic chemistry

ORGANIC CHEMISTRY

The study of HYDROCARBONS and their derivatives.



compounds of C & H



CHEMISTRY

```
graph TD; CHEMISTRY[CHEMISTRY] --> PHYSICAL[PHYSICAL]; CHEMISTRY --> INORGANIC[INORGANIC]; CHEMISTRY --> ORGANIC[ORGANIC]; PHYSICAL --> P_DESC["Law's and application<br/>Physical Phenomenon"]; INORGANIC --> I_DESC["Periodic table<br/>Study of elements and<br/>their properties"]; ORGANIC --> O_DESC["Study of<br/>Carbon compounds"]
```

PHYSICAL

Law's and
application
Physical
Phenomenon

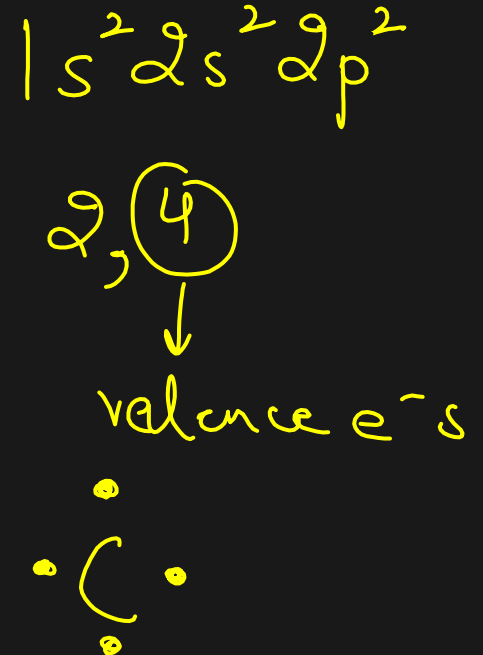
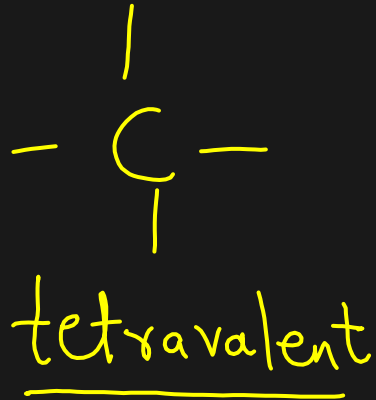
INORGANIC

Periodic table
Study of
elements and
their properties

ORGANIC

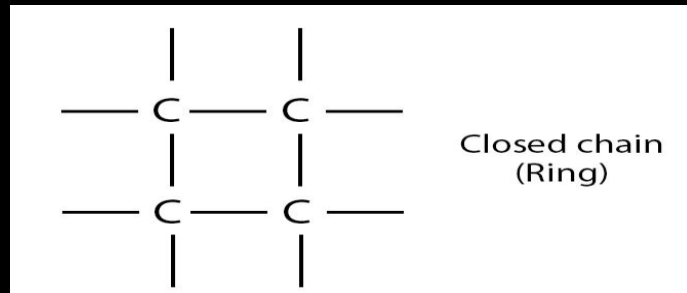
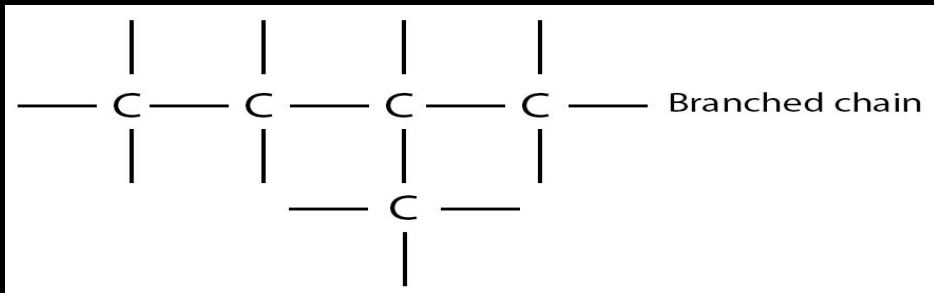
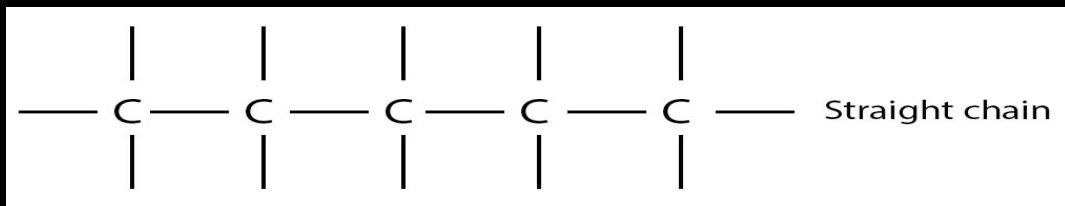
Study of
Carbon
compounds

CARBON

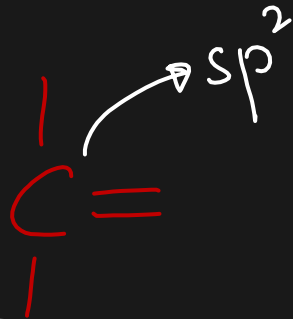
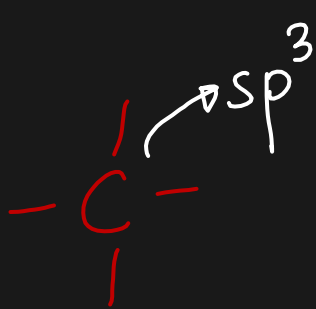


Catenation Nature of Carbon:

Carbon has the unique ability to form bonds with other carbon atoms giving rise to large molecules. This property of carbon is known as catenation. These C – C bonds are very strong and hence, compounds are very stable.

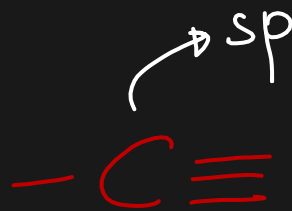


HYBRIDISATION



$$\boxed{\text{Hyb} = sp^{N-1}}$$

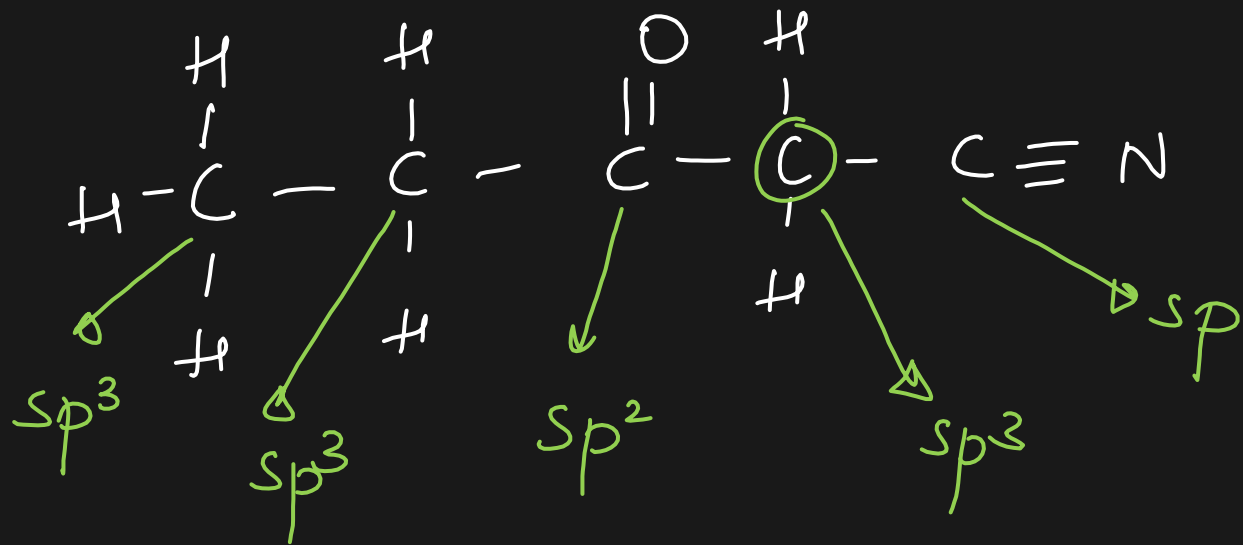
$N = \text{no of } \sigma \text{ bonds} + \text{lone pair}$



sp^3 tetrahedral

sp^2 trigonal planar

sp linear



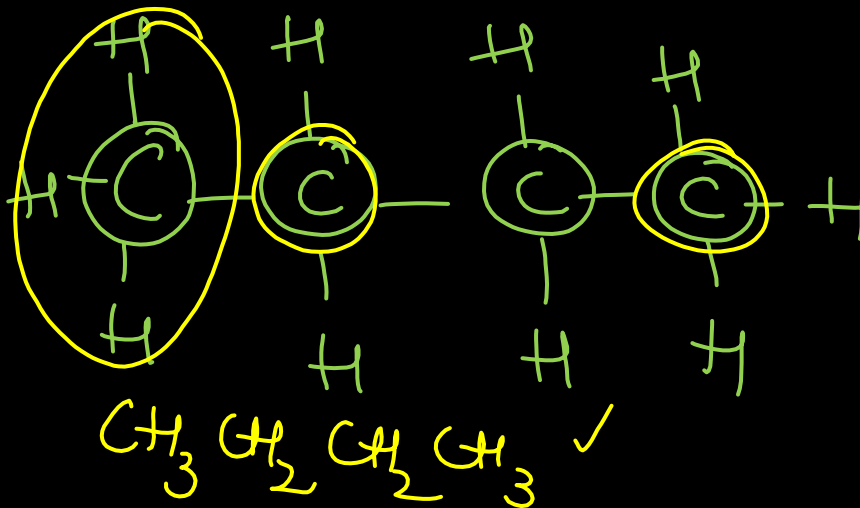
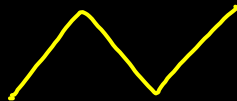
REPRESENTATION OF ORGANIC MOLECULES

Structural Representation of Organic Compounds :

(1) Complete Formula

(2) Condensed Formula

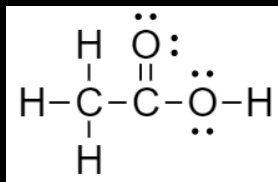
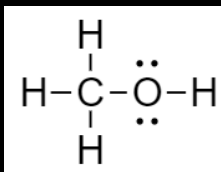
(3) Bond Line Formula



Structural Representation of Organic Compounds :

1) Complete Formula :

It's a representation of a Lewis structure in which bond pairs are represented by A dash (—) and a lone pairs are represented by a pair of dot's (· ·)



Different ways of Writing the Structural Formulae :

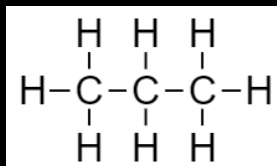
2) Condensed Formula

It's a representation of a molecule in which individual bonds (dashes) are excluded.

Example

i) Propane

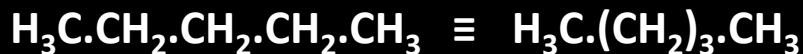
Complete formula

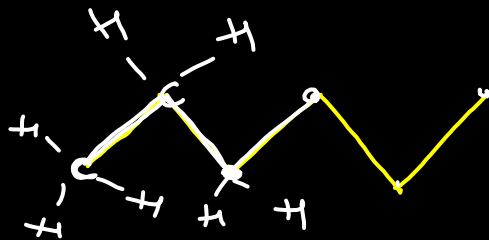
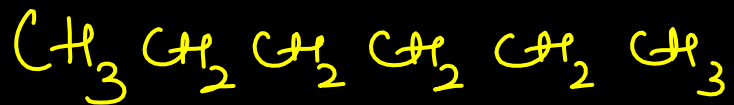
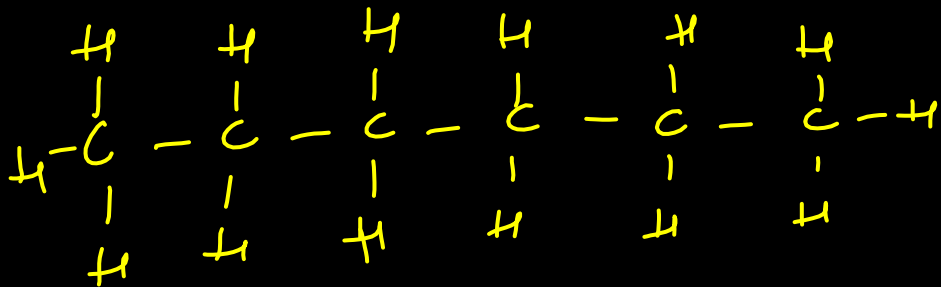


Condensed formula



ii) n-pentane



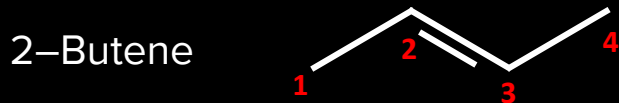
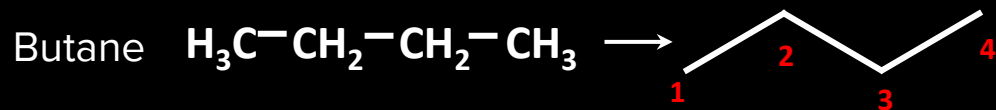


Different ways of Writing the Structural Formulae :

3) Bond Line Formula : [Bond line Or Stick figure]

It's a representation in which bonds are represented by a line and C-atoms are assumed to be present where two lines meet and at the terminals of the line. Also H-atoms are believed to be present on the carbon, however not represented, depending upon the requirement in order to fulfill the valencies.

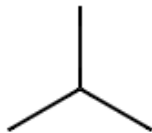
Example



Different ways of Writing the Structural Formulae :

Some more examples

isobutane



Cyclopropane

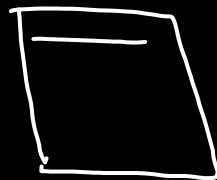
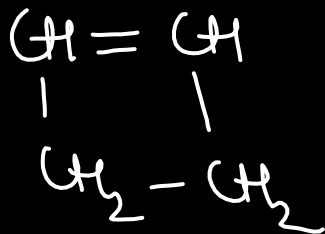
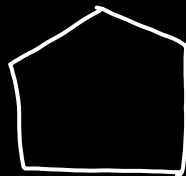
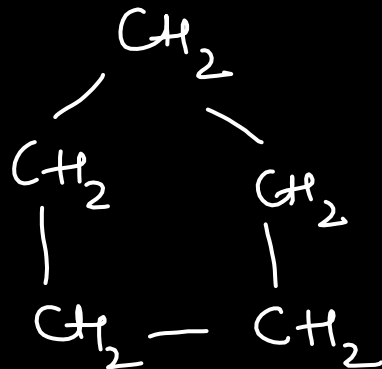
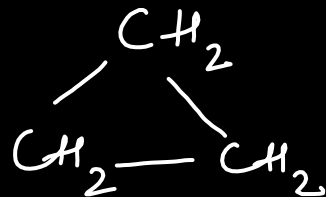
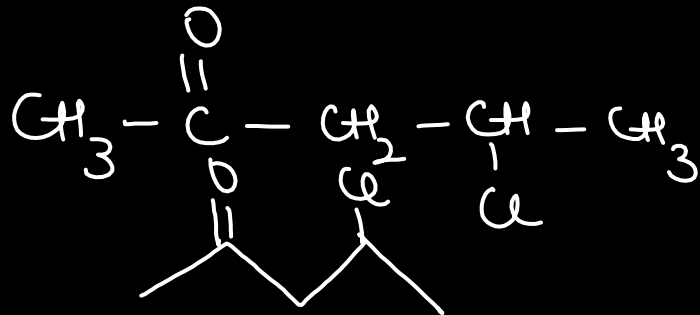
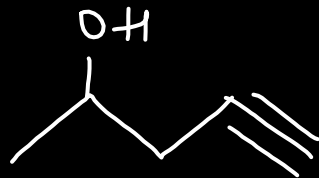
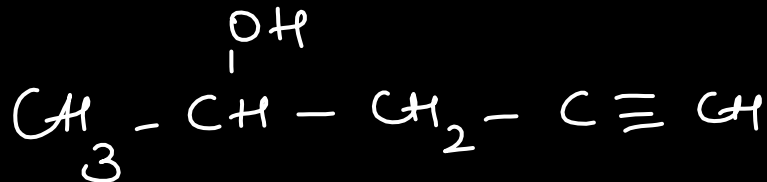
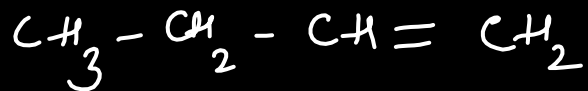


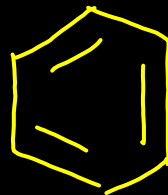
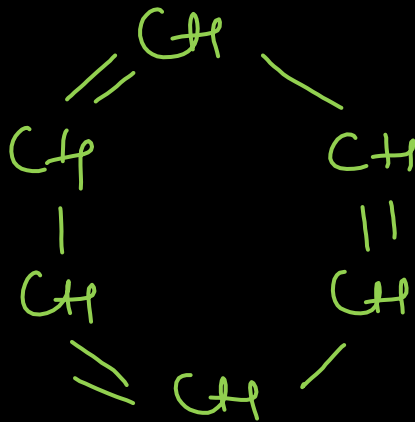
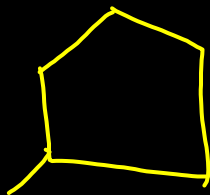
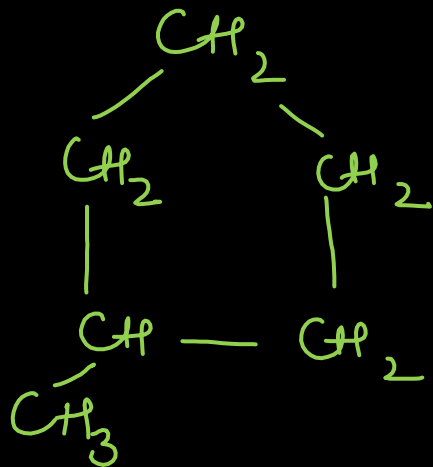
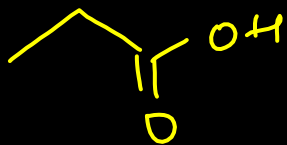
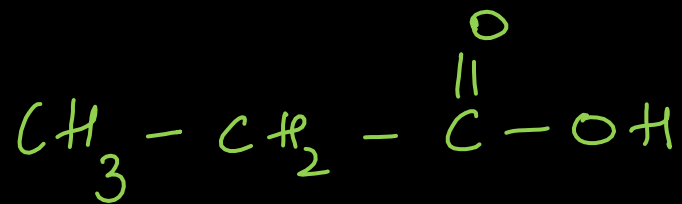
Cyclobutane

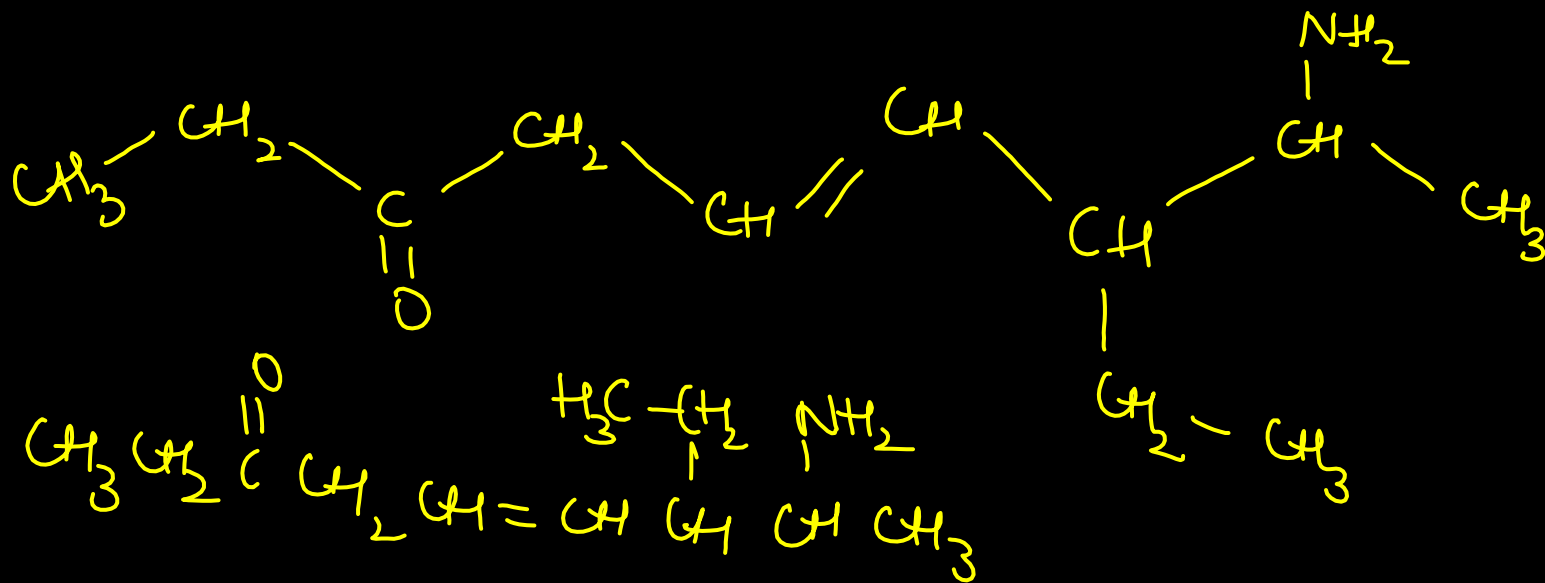
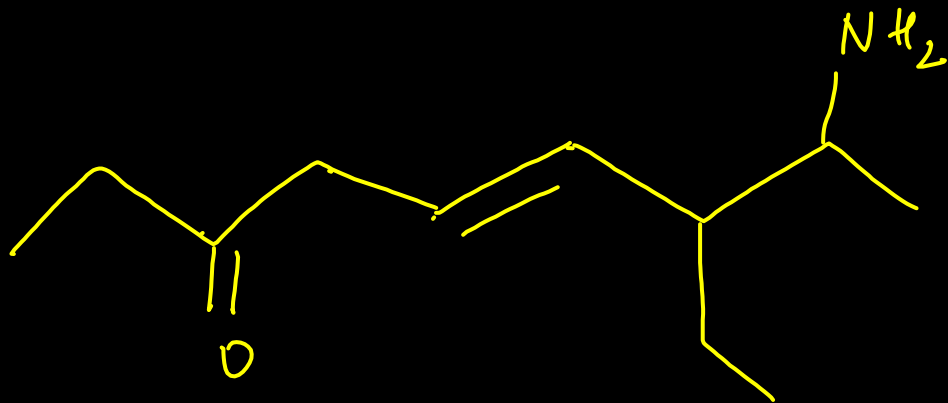


Cyclopentane



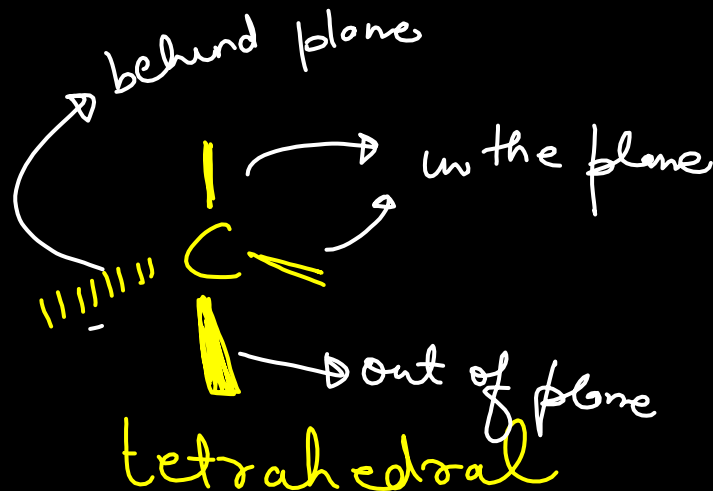
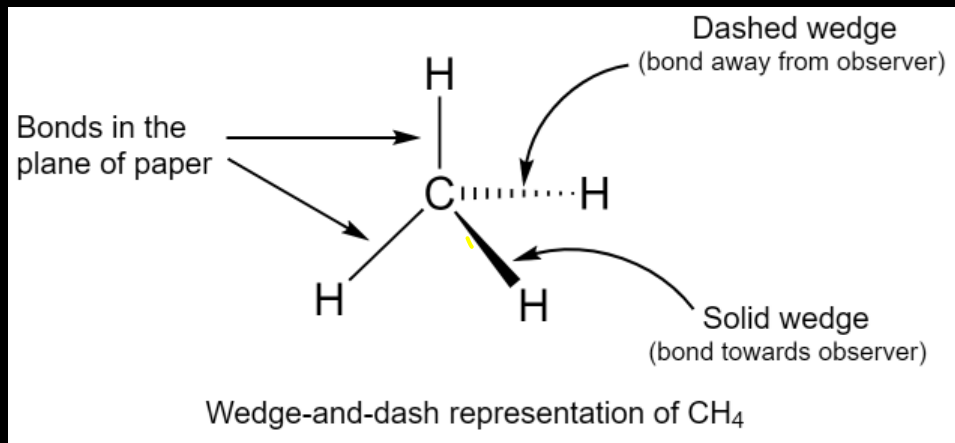






Three-Dimensional Representation of Organic Molecules

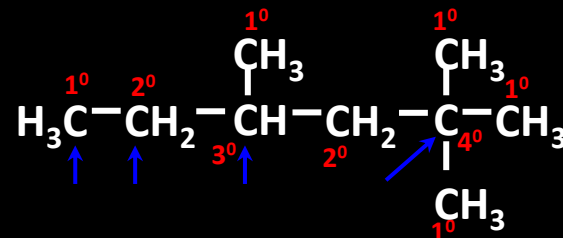
The three-dimensional (3-D) structure of organic molecules can be represented on paper by using certain conventions. For example, by using solid and dashed wedge formula, the 3-D image of a molecule from a two-dimensional picture can be perceived.



Types/Nature of C-atoms in a Molecule

Primary (1°) Carbon atom :

C-atom directly attached to one other C-atom called Primary (1°) C-atom.



Secondary (2°) Carbon atom :

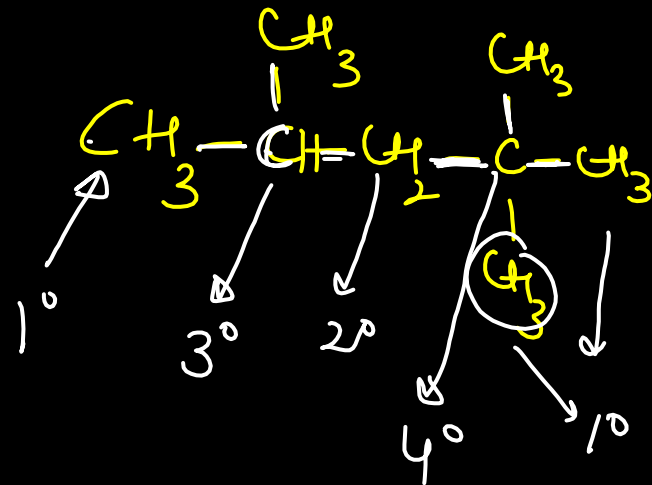
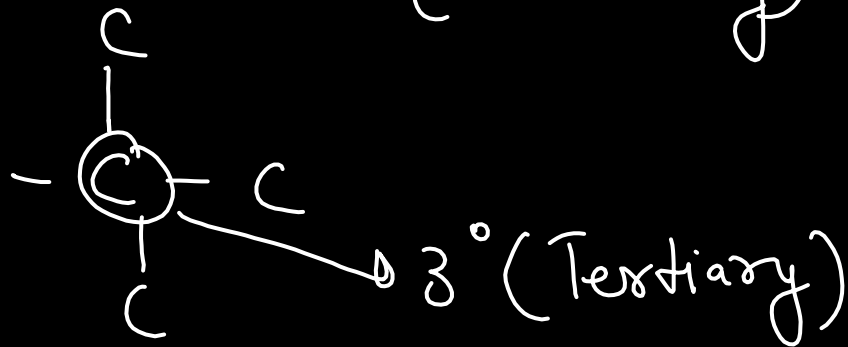
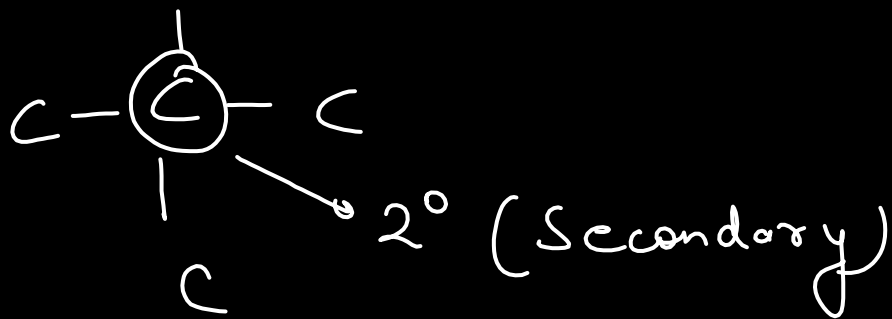
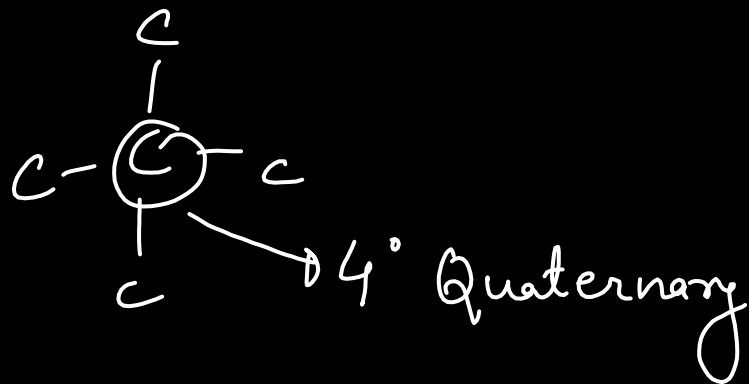
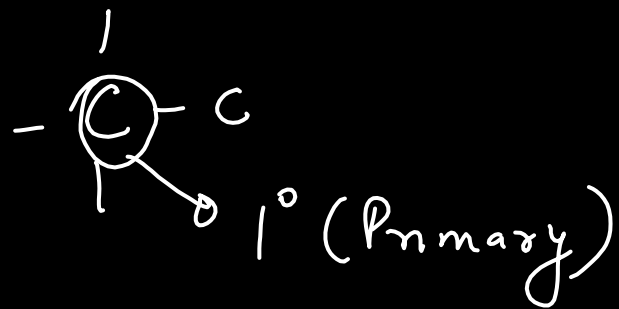
C-atom directly attached to two other C-atoms called Secondary (2°) C-atom.

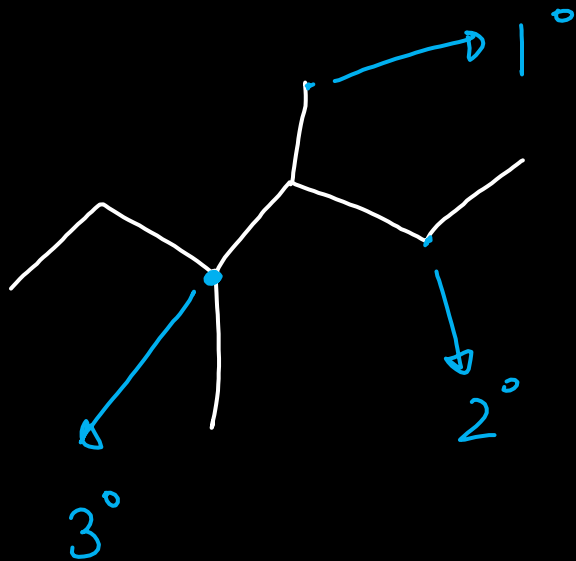
Tertiary (3°) Carbon atom :

C-atom directly attached to three C-atoms called Tertiary (3°) C-atom.

Quaternary (4°) Carbon atom :

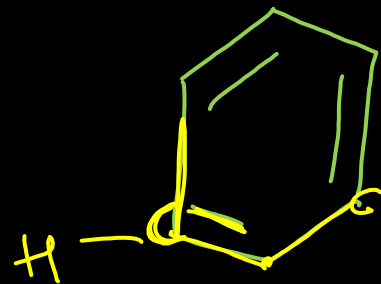
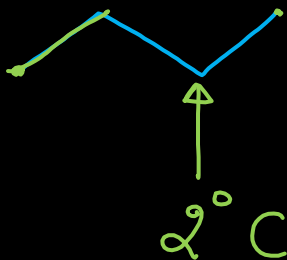
C-atom directly attached to four other C-atoms called Quaternary (4°) C-atom





1° C
 1° H

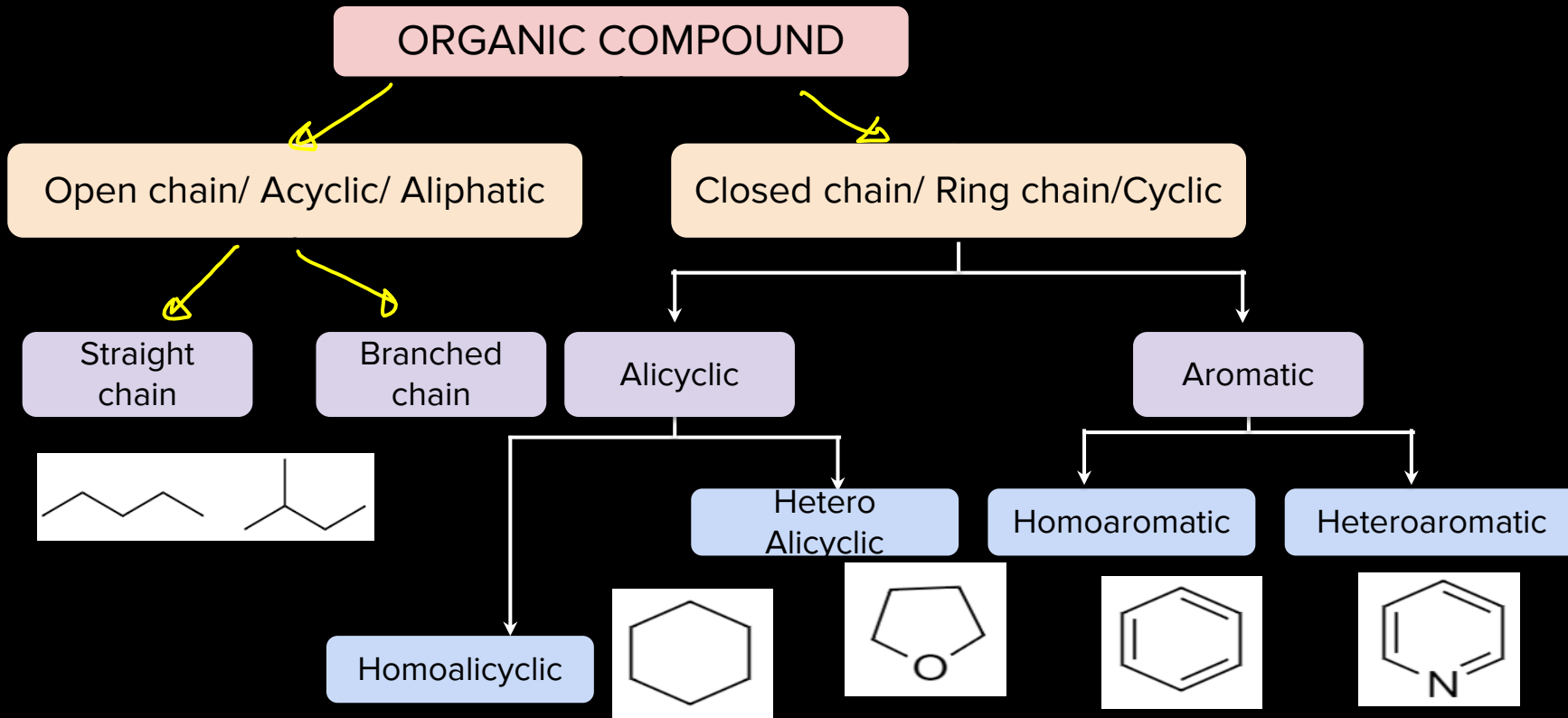
(2)
 (6)

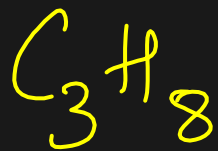
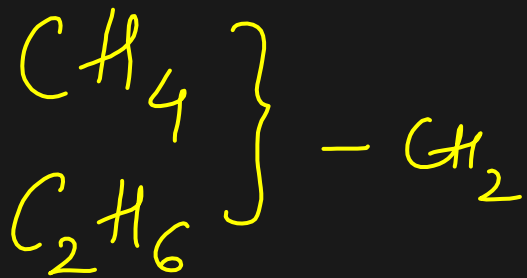


2° C's
 All 6

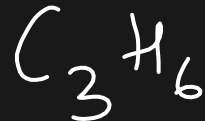
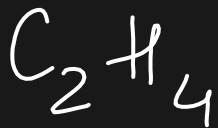
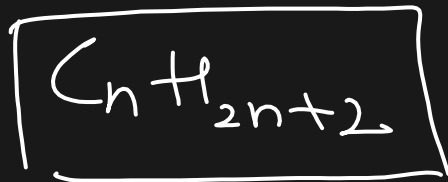
(6) $\rightarrow$ 2° H's

Classification of ORGANIC COMPOUNDS





Alkane



alkene

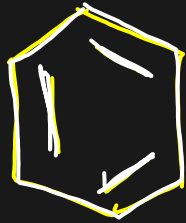
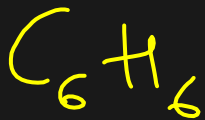


Degree of unsaturation DOU

ring / double bond



$$DOU = \frac{2C + 2 + N - H - X}{2}$$



$$DOU = \frac{2(6) + 2 + 0 - 6 - 0}{2} = 4$$

NOMENCLATURE

IUPAC

Rules

Hydrocarbons

↓ ↓

Saturated Unsaturated

$C-C$ $C=C$
 $C\equiv C$

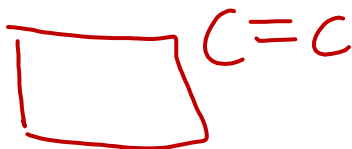
Characteristics of Homologous series

All the members of a homologous series can be represented by the same general formula.

Alkane series



Alkene series



Alkyne series



Alcohol series



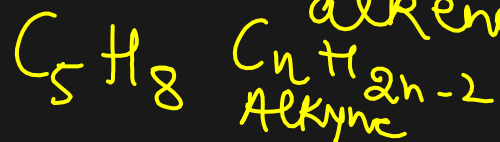
$$n = 6$$



Alkane



alkene



Alkyne

Characteristics of Homologous series

- All the members of a homologous series can be represented by the same general formula.
- Any two adjacent homologues differ by -CH_2 unit.
- The difference in molecular masses of any two adjacent homologues is 14 u.
- All the compounds in a homologous series show same chemical properties.
- All compounds in a homologous series show a gradual change in their physical properties with increase in molecular mass.

International Union of Pure and Applied Chemistry

The official names of the organic compounds were given by:

“International Union of Pure and Applied Chemistry” in 1958.



IUPAC (In short)

These official names are called IUPAC names/nomenclature.



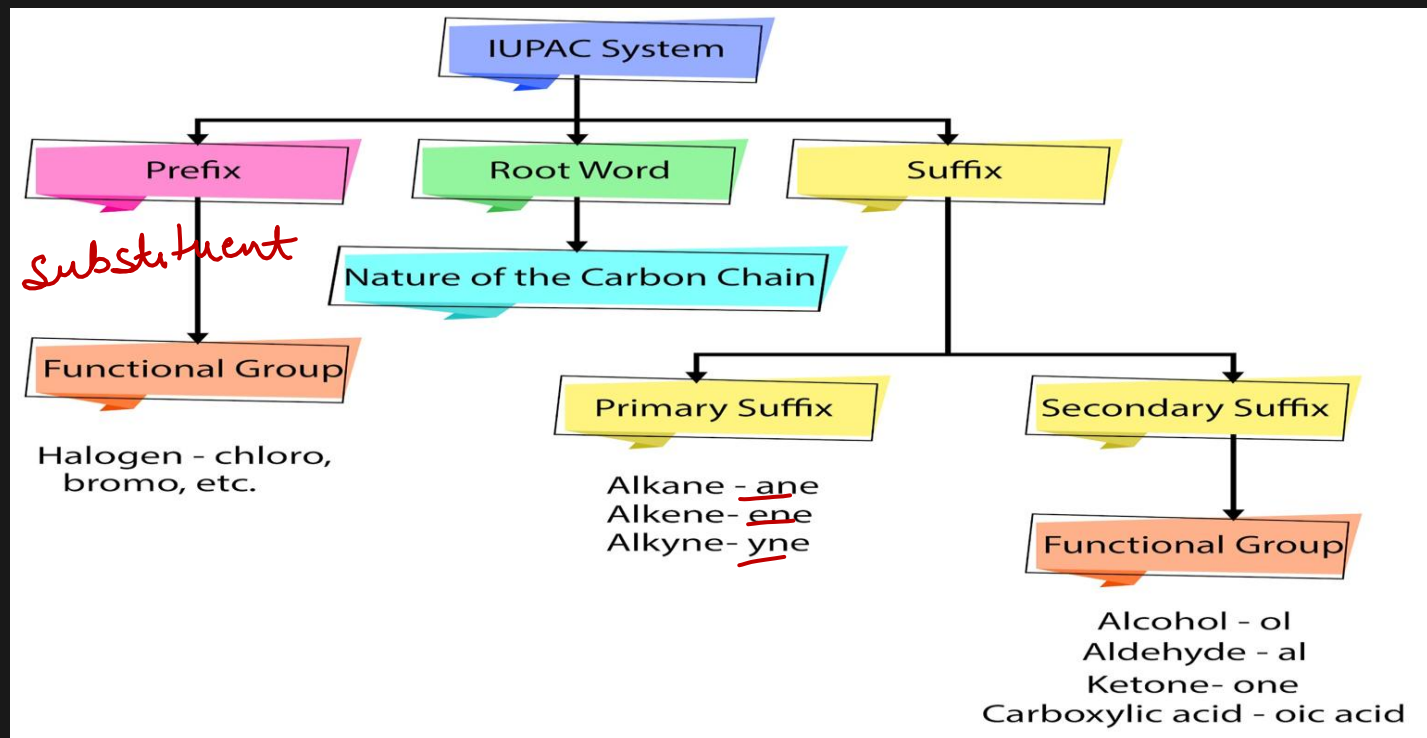
Nomenclature of organic Compounds :

The term 'nomenclature' means the system of naming of organic compounds.

- Trivial or Common system : Names were assigned based on their source of origin or certain properties.
- IUPAC system of nomenclature(Official name)



International Union of Pure and Applied Chemistry



The name of the organic compound according to IUPAC system consists of

Prefix + Word root + Primary suffix + Secondary suffix

IUPAC : Straight Chain Hydrocarbons

The names of straight chain hydrocarbons consist of word root and primary suffix.

Number of Carbon Atoms	Word root
C₁	Meth
C₂	Eth
C₃	Prop
C₄	But
C₅	Pent
C₆	Hex

Number of Carbon Atoms	Word root
C₇	Hept
C₈	Oct
C₉	Non
C₁₀	Dec
C₁₁	Undec
C₁₂	Dodec

IUPAC : Straight Chain Hydrocarbons

C10: dec- (decane)

C20: icos- (icosane)

C30: triacont- (triacontane)

C40: tetracont- (tetracontane)

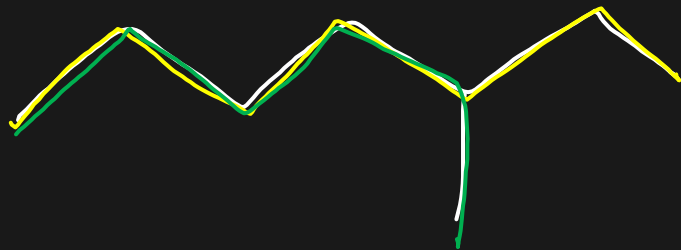
C50: pentacont- (pentacontane)

C60: hexacont- (hexacontane)

Nomenclature of Branched Chain Alkanes:

Certain rules the branched chain hydrocarbons can be named without any difficulty.

- (i) Longest Chain Rule
- (ii) Lowest Number Rule
- (iii) Alphabetical Order of the Side Chain
- (iv) Numbering of Different Alkyl Groups at Equivalent Positions
- (v) Naming the Hydrocarbon with Complex Substituent
- (vi) First point of difference Rule



Butane



Octane



→ heptane

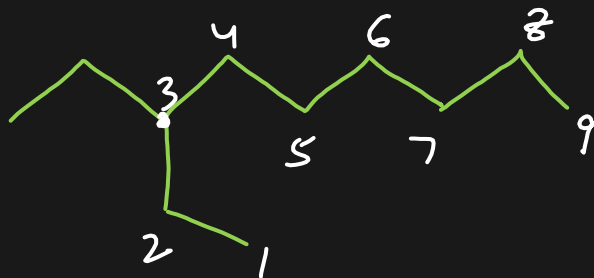


③ Methyl heptane



4-Ethyl octane

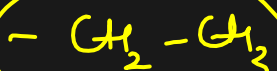
Prefix



3-Ethyl nonane

$-CH_3$
methyl

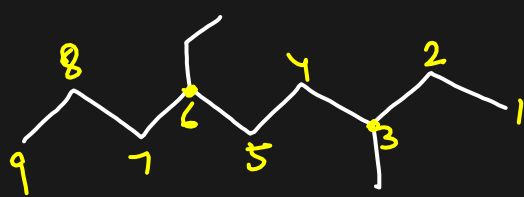
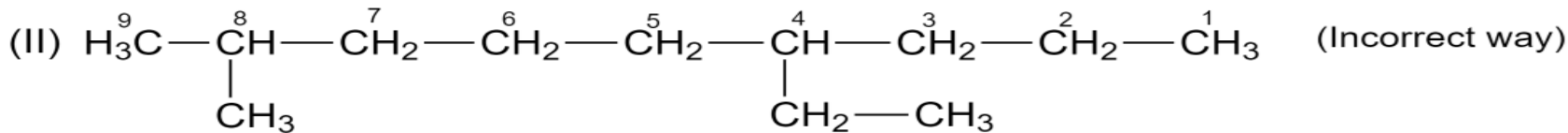
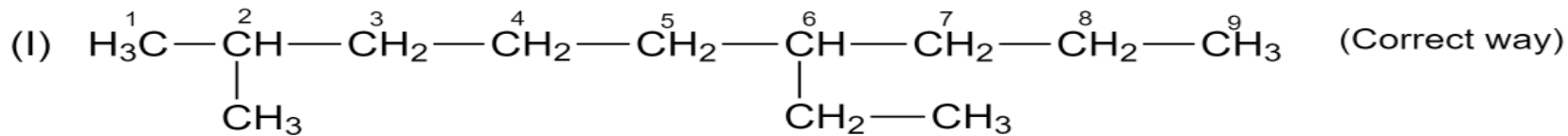
$-C_2H_5$
ethyl



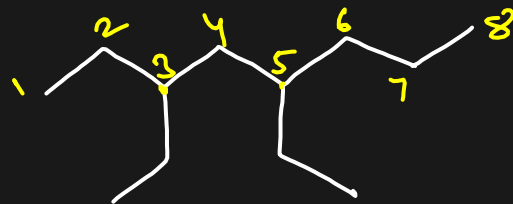
$-C_3H_7$
propyl



3,7-Dimethyl nonane



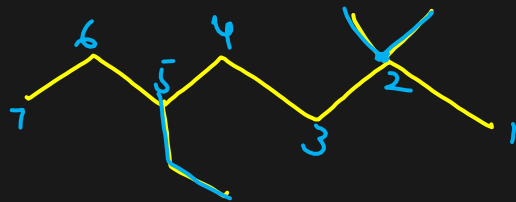
3, 6 ✓
4, 7



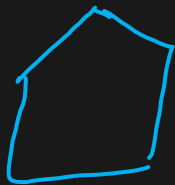
6- Ethyl - 3- methyl nonane

3,5-Diethyloctane

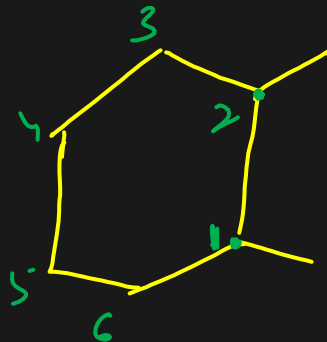
2, 2, 6



5-Ethyl-2,2-dimethyl
heptane



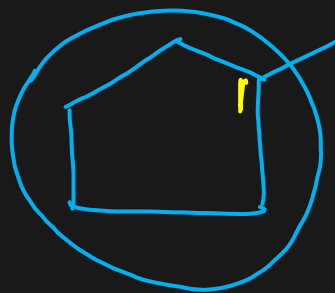
cyclopentane



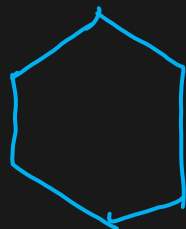
1,2-Dimethylcyclohexane



cyclopropane



1-Methylcyclopentane



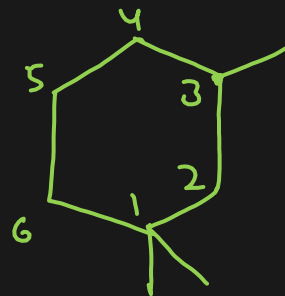
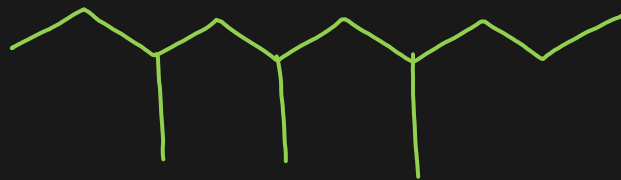
cyclohexane



1-Ethylcyclobutane

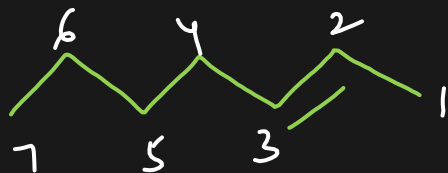


4-Ethyl-5-methylnonane



1,1,3-Trimethylcyclohexane

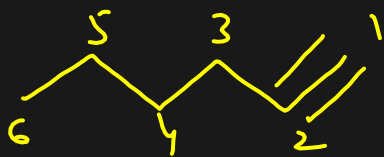
UNSATURATED HYDROCARBONS



Hept-2-ene



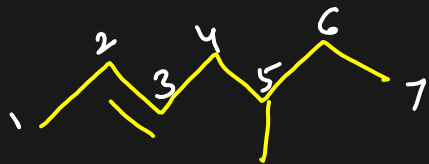
Oct-3-yne



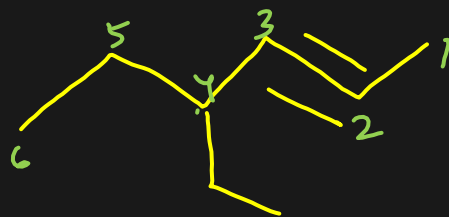
Hex-1-yne



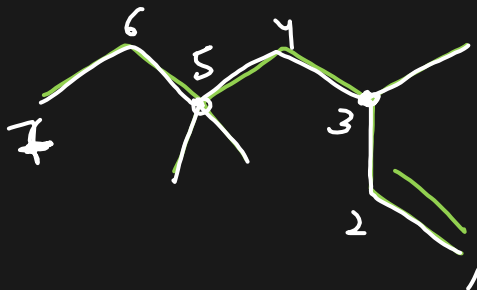
But-2-ene



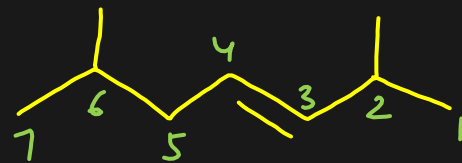
5-Methyl hept-2-ene



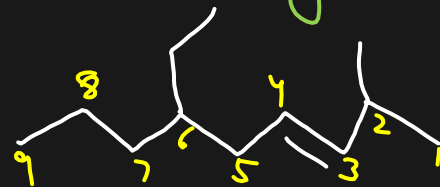
4-Ethyl hex-2-yne



3,5,5-Trimethyl hept-1-ene



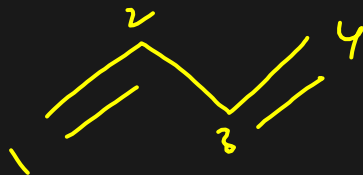
2,6-Dimethyl hept-3-ene



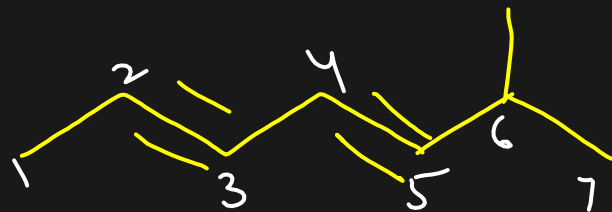
6-Ethyl-2-methyl non-3-ene



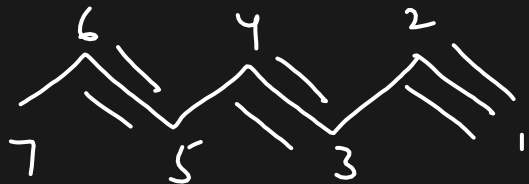
Hexa-1,3-diene



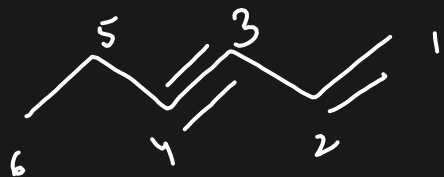
Buta-1,3-diene



6-Methyl hepta-2,4-diene



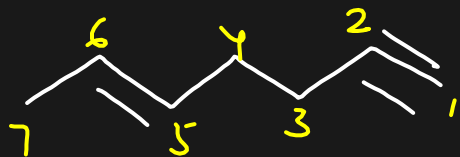
Hepta-1,3,5-triene



Hex-1-en-3-yne



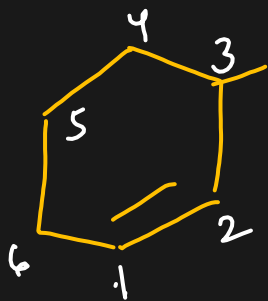
Oct-1-en-7-yne



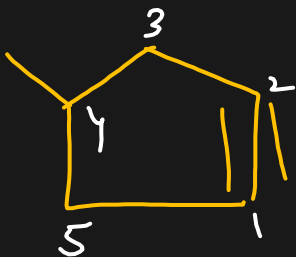
Hept-5-en-1-yne



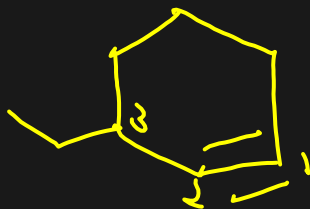
6-Methylhept-4-en-2-yne



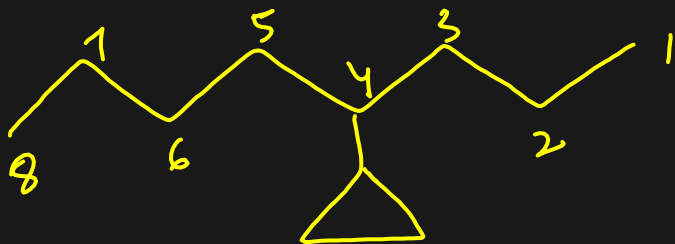
3-Methyl cyclohex-1-ene



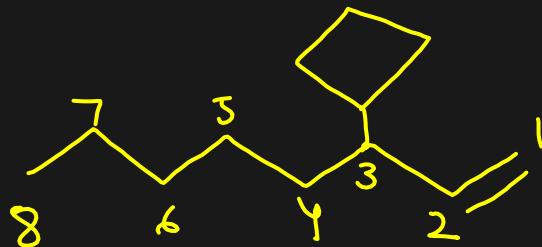
4-Methyl cyclopent-1-yne



3-Ethyl cyclohex-1-yne



⑨ cyclopropyl octane



3-Cyclobutyl oct-1-ene

Functional group

It is an atom or a group of atoms which characterizes the chemical reactivity of a molecule

Class	Formula	Prefix	Suffix
Acid Halide	$-\text{COX}$ $-(\text{C})\text{OX}$	Halo-carbonyl	$-\text{Carbonyl halide}$ $-\text{oyl halide}$
✓ Alcohols	$-\text{OH}$	<u>Hydroxy</u>	$-\text{ol}$
↪ Aldehydes	$-\text{CHO}$ $-(\text{C})\text{HO}$	Formyl Oxo	$-\text{carbaldehyde}$ $-\text{al}$
Amides	$-\text{CONH}_2$ $-(\text{C})\text{ONH}_2$	Carbamoyl -	$-\text{carboxamide}$ $-\text{amide}$
<u>Amines</u>	<u>$-\text{NH}_2$</u>	Amino	$-\text{amine}$

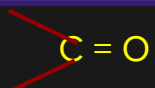
Functional group

It is an atom or a group of atoms which characterizes the chemical reactivity of a molecule

Class	Formula	Prefix	Suffix
Carboxylates	—COO^- —(C)OO^-	Carboxylato -	—Carboxylate —oate
<u>Carboxylic Acid</u>	—COOH —(C)OOH	Carboxy -	—carboxylic acid <u>—oic acid</u>
Ethers	—OR	Alkoxy	—
Esters of Carboxylic Acid	—COOR —(C)OOR	Alkoxy carbonyl —	Alkyl.. Carboxylate Alkyl... oate

Functional group

It is an atom or a group of atoms which characterizes the chemical reactivity of a molecule

Class	Formula	Prefix	Suffix
✓ Ketones	 $\text{C} = \text{O}$	Oxo-	-one
<u>Cyanides</u> (Nitriles)	$-\text{C} \equiv \text{N}$ $-(\text{C}) \equiv \text{N}$	Cyano —	-carbonitrile -nitrile
Salts of Carboxylic	$-\text{COO}^{\ominus} \text{M}^{\oplus}$	—	Cation...carboxylate
Acids	$-(\text{C})\text{OO}^{\ominus} \text{M}^{\oplus}$	—	Cation... oate
Sulphonic Acids	$-\text{SO}_2-\text{OH}$	Sulfo-	-sulfonic acid
Thiols	$-\text{SH}$	Sulfanyl	-thiol

IUPAC naming of functional group containing compounds

✓
Prefix

Substituents

Eg. Methyl

Ethyl

Propyl

Bromo

✓
Compound Root

No. of C in the Parent Chain

Eg. Meth

Eth

Prop

Suffix

Unsaturation Index

Eg. ane

ene

yne

+

Functional Group Suffix (optional)

Eg. ol

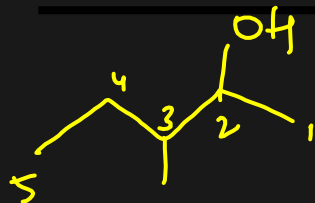
al

oic acid

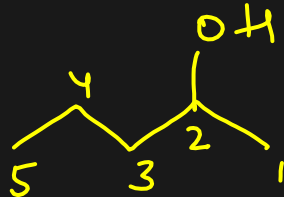
① Alcohol

-OH

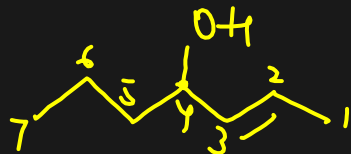
-ol



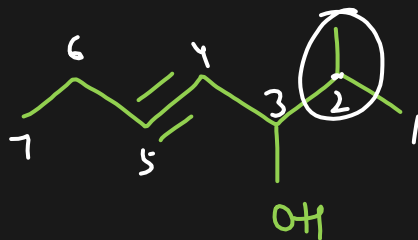
3-Methylpentan-2-ol



Butan-2-ol



Hept-2-en-4-ol



2-Methylhept-4-en-3-ol

② Amine

-NH₂ - amine

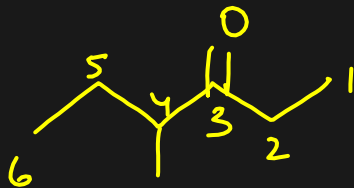
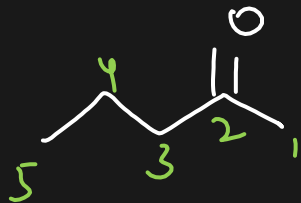
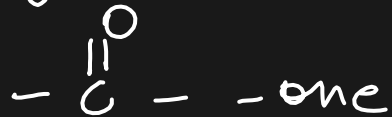


Pentan-2-amine



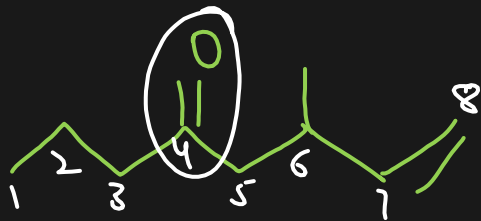
Hex-5-en-2-amine

③ Ketone



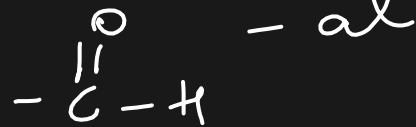
4-methyl hexan-3-one

Pentan-2-one

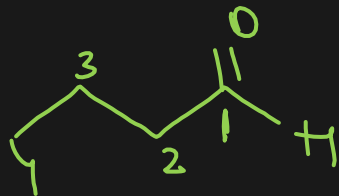


6-Methyl ox-7-en-4-one

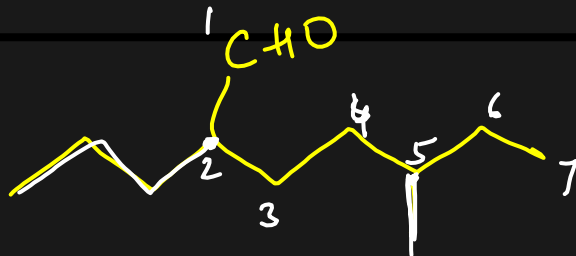
⑨ Aldehyde



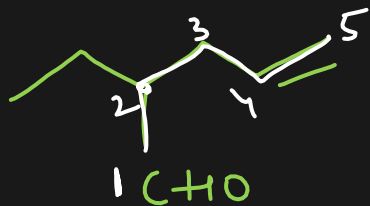
* 1st carbon



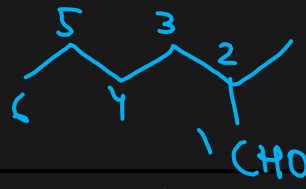
Butan-1-al



5-Methyl-2-propylheptan-1-al



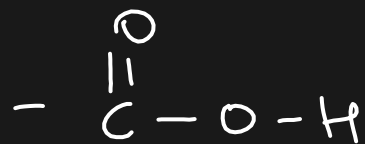
2-Ethylpent-4-en-1-al



2-Methylhexan-1-al

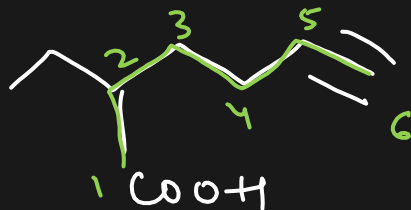
⑤ Carboxylic acid

-COOH



↓
1st Carbon

-oic acid



2-Ethylhex-5-yn-1-oic acid



Hexan-1-oic acid

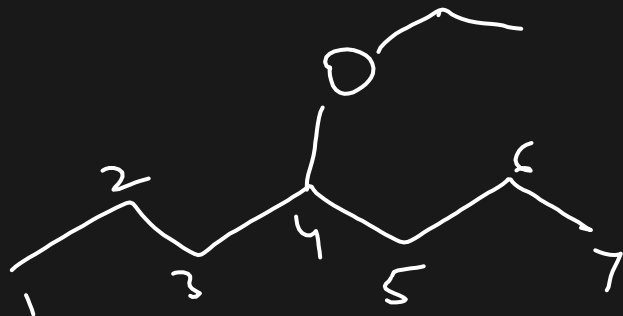
⑥

Ether

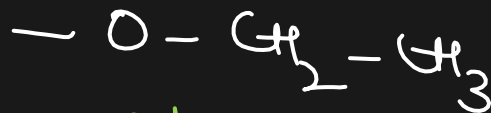
↳ substituent



2-methoxy pentane



4-ethoxy heptane



ethoxy

Esters

↳ fruity smell

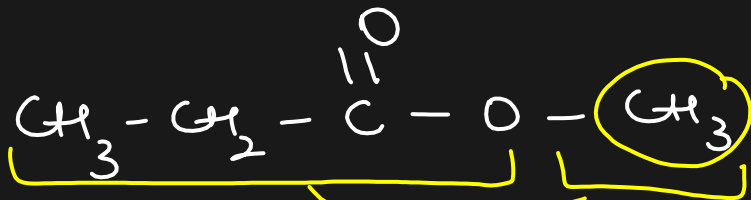


R alkyl

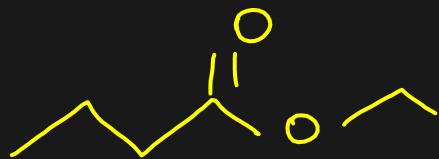
Alkyl alkanoate



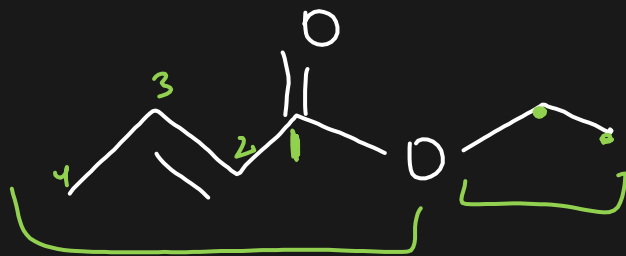
Ethyl ethanoate



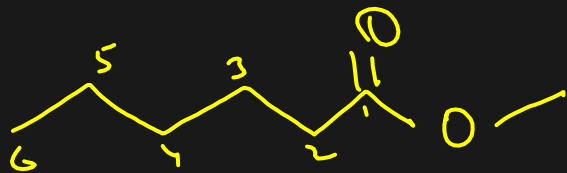
Methyl propanoate



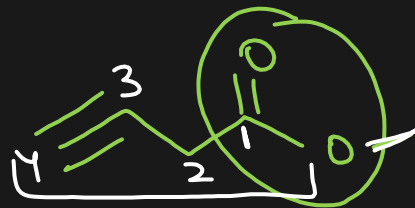
Ethyl butanoate



Ethyl but-2-enoate



Methyl hexanoate



Methyl but-3-ynoate

Order of priority for functional group

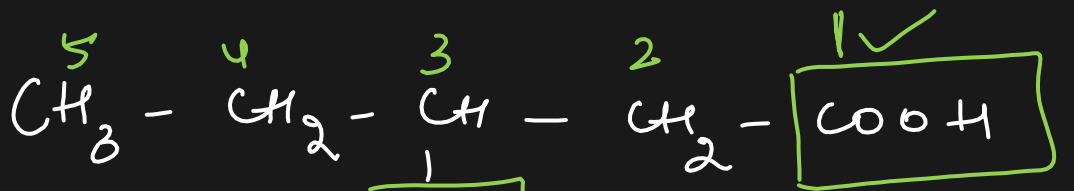
Class of Organic Compound	Functional group
Carboxylic acid	-COOH
Sulphonic acid	-SO ₃ H
Esters	-COOR
Acyl halides	-COX
Amides	-CONH ₂
Nitriles	-CN
Aldehydes	-CHO



Order of priority for functional group

Class of Organic Compound	Functional group
Ketones	$>\text{C}=\text{O}$
Alcohols	$-\text{OH}$
Thiols	$-\text{SH}$
Amines	$-\text{NH}_2$
Alkenes	$>\text{C}=\text{C}<$
Alkynes	$-\text{C} \equiv \text{C}-$





$\swarrow$
 substituent

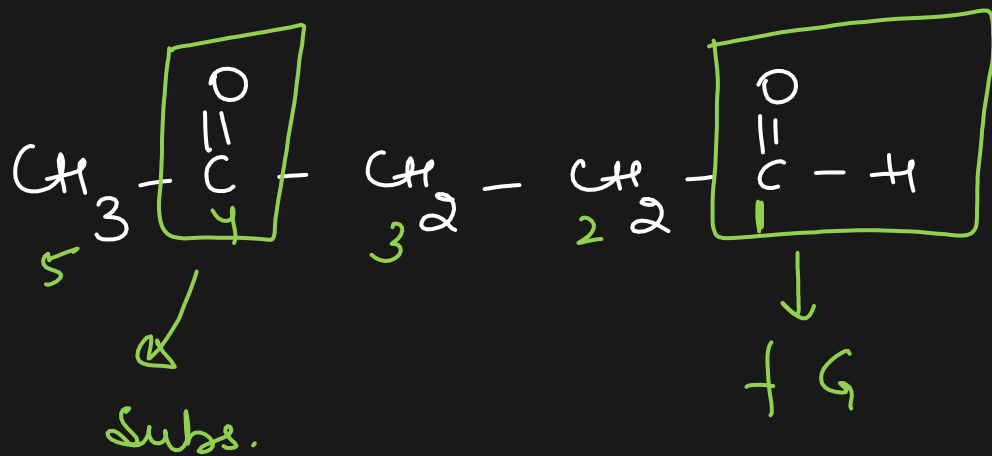
$\downarrow$
 prefix

func gr

$\downarrow$
 suffix

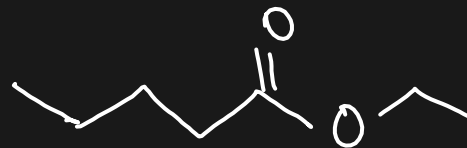
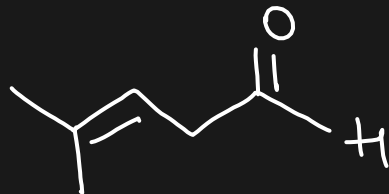
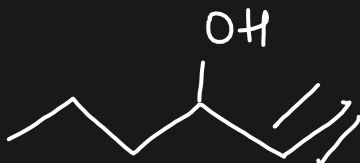
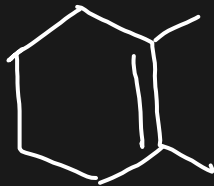
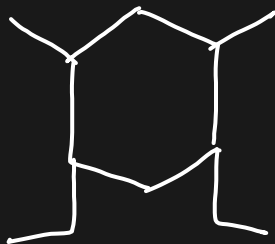
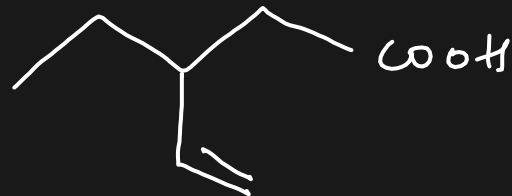
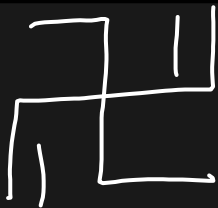
3-hydroxy pentan-1-oic acid

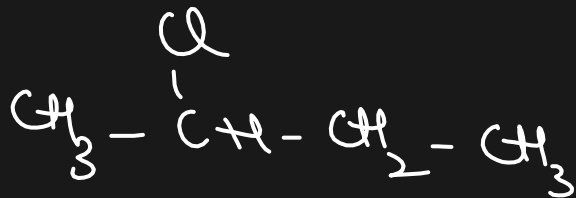
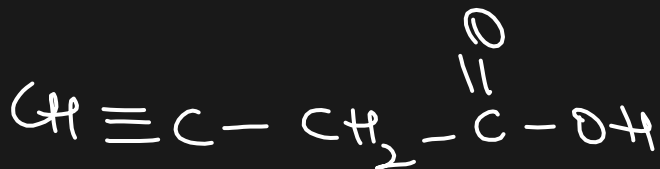
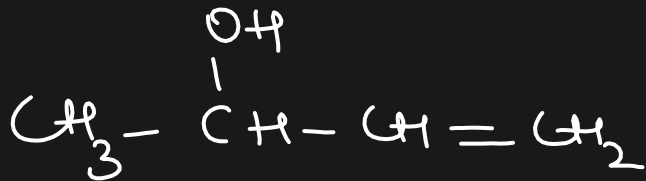
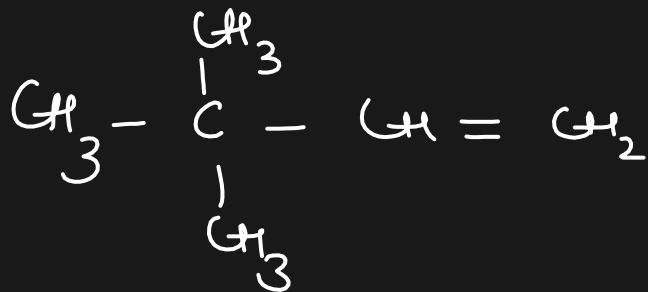
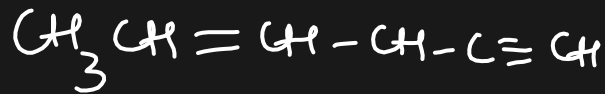
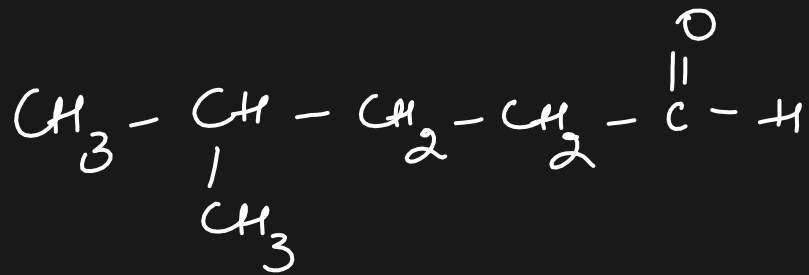
oxo



4-oxopentanal

Home Work





 **Don't Miss Out!**



SUBSCRIBED

Hit **LIKE**, **SHARE**, and **SUBSCRIBE**
for more amazing sessions!

